

Forests for Differences: small panels, the pre-trend diagnostic, and already-treated contamination

Simulation report, second interim instalment

20 August 2026

Abstract

This report covers the three parts of the revision grid that the first interim report could not: the small-panel end of the B2 sample-size sweeps at $N = 50$ and $N = 100$, the operating characteristics of the proposed pre-trend diagnostic on the ten PT_* designs, and the behaviour of every estimator in the suite as the Goodman–Bacon weight on already-treated comparisons is raised from 0.077 to 0.250 on the D_ramp_* designs. It also documents three data-integrity corrections applied to the aggregation layer before any number below was computed.

The small-panel results are favourable to the plain structured DiD-BCF fit on point estimation and unfavourable to it on inference. At $N = 50$ its pooled ATT RMSE is 0.237 in the ordinary sweep, against 0.363 for synthetic DiD and 0.459 for the posterior-corrected fit, and its individual-level CATT surface RMSE is 0.452 against 1.09 to 1.31 for every cohort-time method; but its posterior spread is only 0.54 to 0.70 of the Monte-Carlo error dispersion, and in the serial sweep its ATT coverage sits at 0.83 at every panel size. The posterior correction repairs exactly that: its calibration ratio is close to one at every size and its coverage is at or above nominal throughout, at a roughly doubled RMSE. Callaway–Sant’Anna is the only estimator that fails to return an estimate at $N = 50$, losing four replications in a hundred at $d = 2$ and $d = 3$.

The pre-trend diagnostic is correctly sized (0.045 against a nominal 0.05 when conditional parallel trends holds) and dominates the standard-practice TWFE event-study placebo in power at every violation magnitude, most sharply at the smallest one, where it detects 0.220 against 0.100. It is also the only pre-test in the comparison that stays quiet when conditional parallel trends holds but unconditional parallel trends fails, rejecting 0.080 where the marginal event study and Gardner’s two-stage pre-test reject about 0.75. The power advantage over Callaway–Sant’Anna’s and grf-DiD’s own conditional pre-tests is real but much smaller, so the honest claim is that the diagnostic matches the best available conditional pre-test and beats the marginal one, not that it is unique.

The contamination ramp gives the clearest negative result in the suite. TWFE’s pooled ATT bias rises from 0.261 to 1.181 and its coverage falls from 0.885 to 0.000 across the weight axis, as intended. But DiD-BCF also degrades, from 0.003 to 0.247 (plain) and 0.007 to 0.149 (corrected), while Callaway–Sant’Anna, Gardner and grf-DiD stay flat near 0.01 to 0.03. DiD-BCF is therefore much more robust than TWFE to already-treated contamination and clearly less robust than the estimators built to eliminate it.

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1 Scope and what is new

The first interim report analysed the complete structured grid over the B1, B2 and D designs at panel sizes $N \in \{200, 400, 800, 1600\}$. Three things have changed since, and this report is confined to them.

First, the sample-size sweep was redefined. The live grid is $N \in \{50, 100, 200, 400, 800\}$: the $N = 1600$ column was dropped and two smaller panels were added below the base size, so the sweep now brackets the base panel of 200 on both sides instead of only extending above it. The two new columns were produced locally rather than on Colab, because at these sizes none of the five benchmark estimators is slow.

Second, the ten PT_* designs have completed at 200 replications and at linearity degrees $d = 1$ and $d = 3$, for the DiD-BCF diagnostic and for the pre-tests native to Callaway–Sant’Anna, Gardner, synthetic DiD, grf-DiD and DoubleML, plus a TWFE event-study placebo run on the same replications. The first report contained a 24-replication fragment of three of these designs at $d = 1$ only.

Third, the six D_ramp_nt* designs have completed at 200 replications for every estimator, so estimator error can now be plotted against a measured Goodman–Bacon weight rather than against a design knob that does not move that weight.

Section 2 documents corrections made to the aggregation layer before any number in this report was computed; readers comparing against earlier drafts should read it first, because two of the three corrections change published figures.

2 Data-integrity corrections applied in this pass

Three defects in the result-collection layer were found and fixed. All numbers in this report were computed after the fixes.

The sample-size label on three benchmark estimators. The Callaway–Sant’Anna, Gardner and synthetic-DiD summary files for the sweep panels at $N \in \{400, 800, 1600\}$ carried the value 200 in their N column. The panels themselves were correct, with 400, 800 and 1600 distinct units respectively; only the recorded label was wrong, a residue of an older version of the R driver that hard-coded the base size instead of reading it off the data. Because the aggregation layer

de-duplicates on the key (setting, degree, N , replication, estimand, method), the mislabelled rows collided with the genuine $N = 200$ rows and were silently discarded. The consequence was that these three estimators appeared in the aggregated tables at $N = 200$ only, and the first report's statement that the sweep was complete for them was therefore incorrect. The current R drivers compute N from the panel, so the affected cells were simply re-run; all twelve re-runs completed.

The pre-trend results were not being collected. The aggregation layer globbed `Results/summaries_pretrend_*.csv`, which holds a 24-replication fragment of three designs at $d = 1$, but not `Pretrend/summaries_pretrend_*.csv`, which holds the completed 200-replication runs of all ten designs at $d = 1$ and $d = 3$. Every pre-trend number in the first draft was therefore computed on roughly one eighth of the available replications. The two sources were verified to agree exactly on their 888 overlapping rows before the second was added, so the correction adds replications without revising any completed one.

The sweep axis was hard-coded in the figure code. Three figure routines drew tick marks at 200, 400, 800 and 1600 regardless of the configured sweep, so the two new panels would have been plotted outside the visible axis. The axis is now read from the configuration.

One further gap is documented rather than fixed. The DiD-BCF fits for the two middle ramp designs, `D_ramp_nt20` and `D_ramp_nt25`, have not been run: the notebooks exist but no summary files do. Every ramp result below therefore shows DiD-BCF at four of the six weights, at 0.077, 0.100, 0.196 and 0.250, and the two interior points are dashes rather than interpolations. The competitor estimators and TWFE are present at all six.

3 Designs

3.1 The small-panel extension of B2

The two B2 sweeps are unchanged in every respect except the panel size. The ordinary sweep is the canonical two-period-block design with independent idiosyncratic errors; the serial sweep is the same design with an $AR(1)$ coefficient of 0.6 on the within-unit error. Both are generated at $N \in \{50, 100, 200, 400, 800\}$ and at linearity degrees $d \in \{1, 2, 3\}$, with 100 replications per cell, four pre-periods and four post-periods. The replications are seeded by index, so the panel handed to the R estimators is byte-identical to the one the DiD-BCF notebooks regenerate in memory, and the comparison at a given $(N, d, \text{replication})$ is a comparison of estimators on one dataset rather than of estimators on independent draws.

At $N = 50$ the design has 50 units observed over 8 periods, of which roughly half are treated from period 5: the cohort-time cell `GATT(4, 4)` is therefore estimated from on the order of twenty-five treated units, which is the regime in which a forest-based prior does most of its work and in which a doubly-robust estimator has the least room to cross-fit.

3.2 The pre-trend designs

Ten canonical designs isolate one question each. `PT_hold` is the baseline design, in which conditional parallel trends holds and every rejection is a false positive, so it measures size. The four `PT_violation_g*` designs put the treated group on a differential linear path of slope 0.05, 0.10, 0.20 and 0.40 outcome units per period, tracing a power curve against violation magnitude. `PT_violation_a20` routes a violation of the same nominal magnitude through the *unobserved* confounder rather than through the group indicator, which no adjustment on observables can remove; the realised slope is 0.207, close by construction to the 0.20 of the matching group-trend design. The three `PT_violation_het*` designs impose a differential slope of $\pm\kappa$ that depends on the observed covariate X_1 and cancels in the aggregate, with $\kappa \in \{0.10, 0.20, 0.40\}$: the

realised aggregate slope is zero to three decimals, so an aggregate pre-test is powerless there by construction and only a subgroup contrast can see the violation. Finally `PT_conditional` is the discrimination case, in which conditional parallel trends holds while unconditional parallel trends fails, because assignment depends on a covariate that also shifts trends. A diagnostic whose job is to test the assumption the estimator actually makes should stay quiet there; a marginal event study, which does not condition, should not.

Each design is run at 200 replications at $N = 200$ and at degrees $d = 1$ and $d = 3$. Two decision rules are reported. The *slope* rule tests the single differential-trend coefficient implied by the estimated pre-period path $\Delta(k)$; the *any-k* rule flags if any individual pre-period $\Delta(k)$ is significant after a Bonferroni scaling. When the violation is a linear differential trend the slope rule is the efficient one, so it is reported as the headline and the any- k rule as the conservative alternative.

3.3 The already-treated contamination ramp

The original ramp swept the `dynamic_ramp` parameter, which changes how *inconsistent* the already-treated comparisons are but leaves their Goodman–Bacon *weight* exactly constant at 0.116, because that weight is a function of the adoption design alone. It therefore could not support any claim about what happens as already-treated comparison weight rises. The current design sweeps the never-treated share instead, over $\{0.40, 0.30, 0.25, 0.20, 0.10, 0.05\}$, holding the effect DGP fixed. Shrinking the clean control pool moves the measured mean weight on “later versus earlier” comparisons from 0.077 to 0.250, and that measured weight is the x -axis of every result in Section 6. Each design is a staggered three-cohort adoption with cohort effect multipliers $(1.0, 1.5, 2.0)$ and a dynamic ramp of 0.4, at $N = 200$ with 200 replications and $d \in \{1, 2, 3\}$.

Because the estimators report different estimands natively, the ramp results are read in three internally consistent panels: a pooled ATT for DiD-BCF, TWFE and synthetic DiD; cohort-time cells for Callaway–Sant’Anna, DoubleML and grf-DiD; and event-study coefficients for Gardner. No estimator is scored on a target it does not estimate.

4 Small panels: $N = 50$ and $N = 100$

4.1 Point estimation

Table 1 reports the cohort-time target $\text{GATT}(4, 4)$ and the pooled ATT at $N \in \{50, 100, 200\}$ and $d = 1$, with $N = 200$ carried as the reference column because it is the base panel of every other scenario in the suite. Figure 1 extends the same quantities across the whole sweep.

Every estimator has small bias at every panel size: the largest absolute $\text{GATT}(4, 4)$ bias anywhere in the $d = 1$ small-panel block is 0.083, for the corrected fit at $N = 50$, and most entries are an order of magnitude below the corresponding RMSE. The small-sample problem in this design is therefore dispersion, not location.

On dispersion the plain structured fit wins, and not narrowly. At $N = 50$ in the ordinary sweep its $\text{GATT}(4, 4)$ RMSE is 0.250, against 0.409 for grf-DiD, 0.455 for Callaway–Sant’Anna, 0.707 for the corrected fit and 0.729 for DoubleML. At $N = 100$ the ordering is unchanged, at 0.168 against 0.298, 0.290, 0.342 and 0.467. The same holds on the pooled ATT, where at $N = 50$ the plain fit’s RMSE of 0.237 compares with 0.363 for synthetic DiD and 0.459 for the corrected fit. The advantage narrows as N grows but does not vanish: at $N = 800$ the plain ATT RMSE is 0.051 against 0.092 for synthetic DiD. This is the expected behaviour of a strong regularising prior in a small sample, and it is the clearest quantitative argument in the suite for the DiD-BCF approach at panel sizes typical of applied work.

Table 1: The small-panel end of the B2 sweeps at linearity degree $d = 1$. Bias, RMSE, 95% coverage and the calibration ratio $\overline{\text{sd}}/\text{SD}$ for the cohort-time target GATT(4, 4) and the pooled ATT.

Target	Estimator	$N = 50$				$N = 100$				$N = 200$			
		Bias	RMSE	Cov.	$\overline{\text{sd}}/\text{SD}$	Bias	RMSE	Cov.	$\overline{\text{sd}}/\text{SD}$	Bias	RMSE	Cov.	$\overline{\text{sd}}/\text{SD}$
<i>B2 ordinary</i>													
GATT(4, 4)	DiD-BCF (plain)	-0.019	0.250	0.97	0.80	0.014	0.168	0.98	1.00	-0.022	0.110	1.00	1.00
	DiD-BCF (corrected)	0.083	0.707	0.94	0.95	0.019	0.342	0.95	1.15	-0.028	0.231	0.95	1.05
	Callaway-Sant'Anna	0.055	0.455	0.94	0.90	0.013	0.290	0.97	1.09	-0.031	0.221	0.98	1.03
	DoubleML DR-DiD	0.045	0.729	0.98	0.97	-0.003	0.467	0.97	1.01	-0.038	0.285	0.96	1.01
	grf-DiD (Wang)	0.009	0.409	0.94	0.76	0.006	0.298	0.89	0.81	-0.036	0.222	0.81	0.73
ATT	DiD-BCF (plain)	-0.012	0.237	0.93	0.70	0.010	0.155	0.94	0.81	0.000	0.098	0.96	0.84
	DiD-BCF (corrected)	0.054	0.459	0.96	1.06	0.017	0.280	0.99	1.11	-0.000	0.179	0.98	1.12
	TWFE	—	—	—	—	—	—	—	—	-0.024	0.179	0.98	1.07
	Synthetic DiD	-0.026	0.363	0.97	0.95	0.000	0.258	0.97	1.05	-0.023	0.169	1.00	1.06
<i>B2 serial</i>													
GATT(4, 4)	DiD-BCF (plain)	-0.045	0.288	0.88	0.64	0.008	0.173	0.93	0.77	-0.012	0.111	0.97	0.77
	DiD-BCF (corrected)	0.030	0.478	0.97	0.91	0.014	0.201	0.99	1.26	-0.024	0.136	0.99	1.10
	Callaway-Sant'Anna	0.016	0.301	0.98	0.89	0.016	0.177	0.98	1.17	-0.025	0.129	0.99	1.10
	DoubleML DR-DiD	0.033	0.498	0.98 [†]	0.99	0.033	0.270	0.99	1.15	-0.018	0.168	0.97	1.08
	grf-DiD (Wang)	-0.005	0.291	0.94	0.76	0.013	0.186	0.93	0.90	-0.032	0.133	0.91	0.81
ATT	DiD-BCF (plain)	-0.051	0.298	0.83	0.54	0.005	0.192	0.80	0.58	0.005	0.126	0.83	0.60
	DiD-BCF (corrected)	0.003	0.389	0.97	0.96	0.013	0.216	0.98	1.17	-0.002	0.150	0.98	1.10
	TWFE	—	—	—	—	—	—	—	—	-0.022	0.198	0.98	1.04
	Synthetic DiD	-0.040	0.357	0.97	0.95	-0.003	0.250	0.96	1.07	-0.024	0.166	0.99	1.08

Note. 100 replications per cell. [†] computed on the replications in which the estimator returned an estimate. TWFE was not run on the sweeps below $N = 200$. Gardner reports event-study coefficients only and synthetic DiD a pooled ATT only, so neither appears against every target.

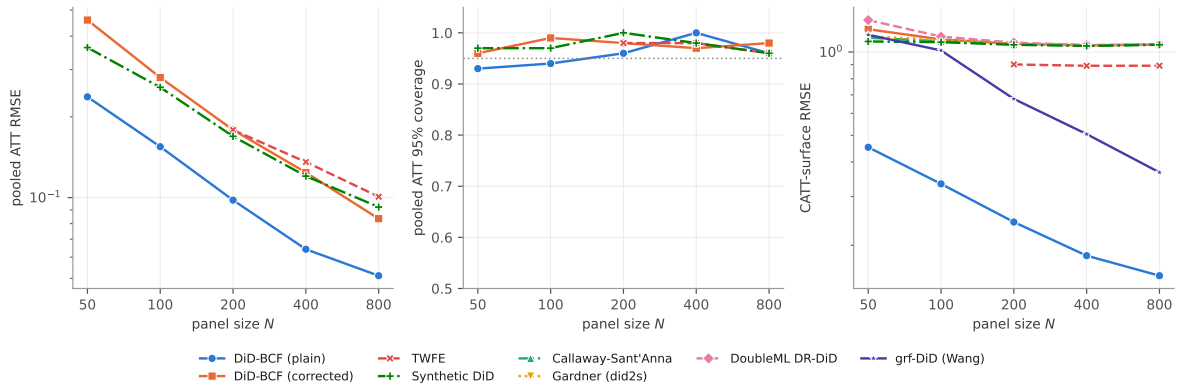


Figure 1: Pooled ATT RMSE, pooled ATT coverage, and individual-level CATT-surface RMSE across the full B2 ordinary sweep at $d = 1$. The two smallest panels are the new columns.

4.2 Inference, and the price of the prior

The same prior that buys the point-estimation advantage is paid for in the interval. The calibration ratio $\overline{\text{sd}}/\text{SD}$, the mean reported posterior spread divided by the Monte-Carlo standard deviation of the error, is the diagnostic: a value below one means the estimator is reporting less uncertainty than it actually has. For the plain fit it is 0.70 at $N = 50$ and 0.84 at $N = 200$ on the ordinary-sweep ATT, and 0.54, 0.58 and 0.60 at $N = 50, 100$ and 200 on the serial-sweep ATT. The serial figures do not improve with N at all, which is what one expects of a specification that does not model the within-unit dependence: the shortfall is a missing variance component, not a small-sample artefact. Coverage follows: the plain serial ATT covers 0.83, 0.80, 0.83, 0.85 and 0.83 at the five panel sizes, flat and roughly twelve points below nominal.

The posterior correction repairs precisely this. Its calibration ratio on the serial ATT is 0.96, 1.17, 1.10, 1.09 and 0.92 across the sweep and its coverage is 0.97 or above at every size, at the cost of an RMSE roughly 1.1 to 1.3 times the plain fit’s on that target, and about twice the plain fit’s on the ordinary sweep. At $d = 1$ this is a clean efficiency-for-honesty trade and the correction is the specification to report when interval coverage matters.

That endorsement does not survive the nonlinear degrees at small N . At $d = 2$ and $N = 50$ the corrected ordinary-sweep GATT(4, 4) RMSE is 2.517, ten times the plain fit’s 0.263 and five times Callaway–Sant’Anna’s 0.462; at $d = 3$ it is 2.455. The instability persists into the middle of the sweep, with a corrected RMSE of 1.319 at $N = 400$ and $d = 2$ against 0.092 for the plain fit, before recovering to 0.318 at $N = 800$. Coverage stays near nominal throughout, so the correction is not producing confidently wrong answers; it is producing very wide ones. The practical reading is that the cross-fitted correction needs either a larger panel or a degree-one nuisance structure to be stable, and that the corrected estimator should not be presented as a general-purpose replacement for the plain fit.

4.3 Estimability at $N = 50$

Callaway–Sant’Anna is the only estimator in the comparison that fails to return an estimate. At $N = 50$ and degrees $d = 2$ and $d = 3$ it returns nothing for any cohort-time cell in four replications out of a hundred in both sweeps, so its retention is 0.96; at $N = 100$ it is 0.99 and at $N \geq 200$ it is 1.00. DoubleML loses one replication at $N = 50$ in the serial sweep at $d = 1$ and two at $d = 3$. Every other estimator, including both DiD-BCF specifications, returns an estimate in every replication at every cell. The metrics layer computes all statistics on the replications that produced an estimate and records the retained fraction alongside, so the affected cells are marked in Table 1 rather than being silently averaged; a coverage of 0.94 over 96 replications is not the same statement as a coverage of 0.94 over 100, and the distinction matters most at exactly the panel size where the comparison is being made.

4.4 The CATT surface

Table 2 reports the individual-level surface: the within-replication RMSE, MAPE and pointwise coverage over the treated observations. Every estimator other than DiD-BCF reports a single number per cohort-time cell, so its surface is that number broadcast to the individuals in the cell, and the comparison measures exactly what is lost by not modelling individual heterogeneity.

The result is the strongest in this report. At $N = 50$ in the ordinary sweep the plain fit’s surface RMSE is 0.452 with a MAPE of 0.113 and pointwise coverage of 0.92; every cohort-time method sits between 1.09 and 1.31 with a MAPE near 0.28 and coverage between 0.50 and 0.75. The gap is present at the smallest panel in the grid, which is the most demanding place to claim it, and it widens with N rather than closing: at $N = 800$ the plain surface RMSE is 0.155 against 1.06 for the cohort-time methods, whose surface error is bounded below by the within-cell heterogeneity they cannot represent and therefore does not converge. grf-DiD is the only benchmark that improves along this axis, from 1.155 at $N = 50$ to 0.367 at $N = 800$,

because it also produces individual-level predictions; it remains between two and three times the plain fit’s error at every size.

Two qualifications belong with that result. The corrected fit does not share it: its surface RMSE is 1.209 at $N = 50$ and 1.064 at $N = 800$, in the cohort-time band rather than the DiD-BCF band, because the correction recentres the posterior at the cell level and discards the individual variation. And the plain fit’s surface advantage coexists with its scalar-interval shortfall documented above, so the honest summary is that the plain specification is the one to use for heterogeneity and the corrected one for pooled inference.

Table 2: Individual-level CATT surface at the small end of the B2 sweeps, $d = 1$. Within-replication RMSE and MAPE over the treated observations, and pointwise 95% coverage, averaged across replications.

Design	Estimator	$N = 50$			$N = 100$			$N = 200$		
		RMSE	MAPE	Cov.	RMSE	MAPE	Cov.	RMSE	MAPE	Cov.
B2 ordinary	DiD-BCF (plain)	0.452	0.113	0.92	0.334	0.083	0.95	0.243	0.058	0.97
	DiD-BCF (corrected)	1.209	0.309	0.67	1.111	0.292	0.48	1.071	0.285	0.32
	TWFE	—	—	—	—	—	—	0.902	0.221	0.35
	Callaway–Sant’Anna	1.126	0.291	0.57	1.094	0.288	0.41	1.069	0.285	0.29
	Gardner (did2s)	1.139	0.285	0.58	1.107	0.289	0.45	1.071	0.284	0.32
	DoubleML DR-DiD	1.305	0.324	0.75	1.137	0.295	0.56	1.082	0.288	0.38
	Synthetic DiD	1.092	0.279	0.50	1.085	0.286	0.37	1.062	0.282	0.26
	grf-DiD (Wang)	1.155	0.296	0.54	1.013	0.268	0.48	0.677	0.166	0.60
B2 serial	DiD-BCF (plain)	0.499	0.124	0.84	0.380	0.095	0.87	0.295	0.072	0.89
	DiD-BCF (corrected)	1.145	0.292	0.58	1.087	0.288	0.40	1.063	0.284	0.26
	TWFE	—	—	—	—	—	—	0.902	0.221	0.35
	Callaway–Sant’Anna	1.102	0.284	0.49	1.079	0.286	0.35	1.063	0.284	0.25
	Gardner (did2s)	1.141	0.283	0.57	1.108	0.289	0.44	1.071	0.284	0.32
	DoubleML DR-DiD	1.197	0.301	0.70	1.120	0.295	0.49	1.073	0.287	0.32
	Synthetic DiD	1.092	0.278	0.50	1.084	0.285	0.37	1.062	0.282	0.25
	grf-DiD (Wang)	1.134	0.290	0.48	0.989	0.263	0.42	0.615	0.150	0.59

Note. Every estimator other than DiD-BCF reports a single number per cohort-time cell, so its surface is that number broadcast to the individuals in the cell; the comparison measures what is lost by not modelling individual heterogeneity.

5 The pre-trend diagnostic

5.1 Size, power and discrimination

Table 3 reports the diagnostic against the TWFE event-study placebo at $d = 1$; Table 4 repeats it at $d = 3$. All figures are over 200 replications, except `PT_violation_het20` at $d = 3$, where one block of 50 replications is missing and the figures are over 150.

Under `PT_hold` the diagnostic rejects 0.045 of the time against a nominal 0.05, with a Monte-Carlo standard error of about 0.015, and the any- k rule rejects 0.030. At $d = 3$ the corresponding figures are 0.055 and 0.045. The diagnostic is correctly sized, and the conservative rule is conservative as intended.

Power rises monotonically with the violation, from 0.220 at a slope of 0.05 through 0.520 at 0.10 and 0.935 at 0.20 to 1.000 at 0.40. The TWFE event-study placebo run on the same replications rejects 0.100, 0.250, 0.700 and 1.000. The advantage is therefore largest where it matters most, at the small violations that a pre-test is actually needed to catch: at a slope of 0.05 the diagnostic more than doubles the placebo’s rejection rate, and at 0.10 it roughly doubles it, while at 0.40 both detect everything. The violation routed through the unobserved confounder behaves like the group-trend violation of the same realised magnitude, at 0.930 against 0.705, confirming that the diagnostic responds to the differential path rather than to the channel that produced it.

Discrimination is the sharpest of the three results. Under `PT_conditional`, where the estimator’s own assumption is satisfied and only unconditional parallel trends fails, the diagnostic rejects 0.080 while the marginal event study rejects 0.755. Standard practice would reject a correctly specified analysis in three cases out of four here. The diagnostic’s slope estimate is also nearly unbiased in that design, at 0.055 against a true zero with 0.950 coverage, whereas the event study’s slope carries a bias of 0.287 and covers 0.245.

Table 3: Operating characteristics of the pre-trend diagnostic at $d = 1$, against the TWFE event-study placebo run on the same replications. 200 replications per scenario.

Scenario	Violation	True slope	Detection, slope rule		Any- k Diagnostic	Slope recovery	
			Diagnostic	TWFE ES		bias	cover. 95%
PT_violation_het40	via α_i	-0.001	0.115	0.065	0.075	0.045	0.94
PT_violation_het20	via α_i	-0.001	0.055	0.055	0.040	0.043	0.94
PT_violation_het10	via α_i	-0.000	0.040	0.045	0.025	0.041	0.94
PT_violation_a20	via α_i	0.207	0.930	0.705	0.845	0.016	0.98
PT_conditional	unconditional only	0.000	0.080	0.755	0.040	0.055	0.95
PT_violation_g05	group slope	0.050	0.220	0.100	0.120	0.031	0.96
PT_violation_g10	group slope	0.100	0.520	0.250	0.400	0.025	0.97
PT_violation_g20	group slope	0.200	0.935	0.700	0.880	0.017	0.96
PT_violation_g40	group slope	0.400	1.000	1.000	1.000	0.010	0.96
PT_hold	none (PTA holds)	0.000	0.045	0.045	0.030	0.040	0.96

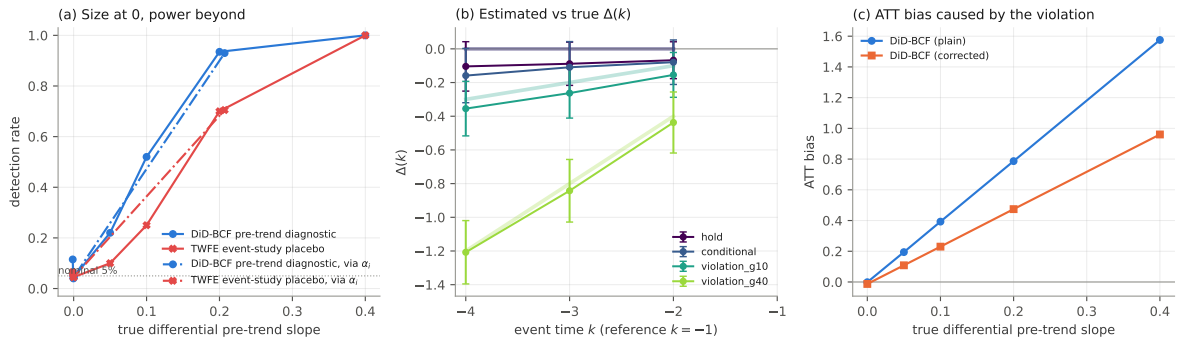


Figure 2: Detection rate against the realised differential pre-trend slope, the estimated $\Delta(k)$ profiles, and the ATT bias the violation induces, at $d = 1$.

5.2 Against the benchmarks’ own pre-tests

Comparing the diagnostic only against a marginal event study overstates the case, because Callaway–Sant’Anna, grf-DiD, Gardner, DoubleML and synthetic DiD all admit pre-tests of their own and three of them condition on covariates. The full comparison is in `Pretrend/BENCHMARK_TABLES.txt` and its qualitative content is as follows.

On the group-trend power curve at $d = 1$ the diagnostic leads, at 0.220, 0.520, 0.935 and 1.000, against 0.090, 0.320, 0.830 and 1.000 for Callaway–Sant’Anna, 0.085, 0.285, 0.795 and 1.000 for grf-DiD, 0.095, 0.240, 0.700 and 1.000 for Gardner, and 0.085, 0.180, 0.540 and 0.985 for DoubleML. The margin over the best conditional competitor is therefore about thirteen points at the smallest violation and twenty at the second smallest, which is a real but modest advantage, not the order-of-magnitude gap the comparison against the marginal event study suggests. Synthetic DiD is a special case: its pre-test rejects 0.000 at slopes up to 0.10 and only 0.170 at 0.20, because the synthetic-control weights absorb the differential trend into the fit, so it is barely a pre-test of the assumption at all.

Two competitors fail the discrimination case in the same way the marginal event study does. Gardner rejects 0.750 and the TWFE placebo 0.755 under `PT_conditional`, because both pre-test an unconditional path; Callaway–Sant’Anna rejects 0.070 and grf-DiD 0.100, close to the diagnostic’s 0.080. The defensible claim is therefore narrower than “uniquely able to discriminate”: the diagnostic, Callaway–Sant’Anna’s pre-test and grf-DiD’s pre-test all correctly stay quiet, and the two unconditional pre-tests do not.

5.3 The heterogeneous violation

The three `PT_violation_het*` designs impose a violation that cancels in the aggregate, so the realised aggregate slope is zero to three decimals. Every aggregate pre-test correctly rejects at about its nominal rate there, between 0.040 and 0.115; that is the right answer, not a failure, and the induced ATT bias confirms it, at -0.005 to -0.001 for the plain fit. The interesting quantity is the subgroup contrast in X_1 , which is the only statistic that can see this violation. The realised contrast slope is 2κ . At a contrast slope of 0.40 the diagnostic’s contrast rejects 0.520, Callaway–Sant’Anna’s 0.745, grf-DiD’s 0.700 and Gardner’s 0.665; at 0.80 all four reach 1.000; at 0.20 the diagnostic rejects 0.020 against 0.320, 0.275 and 0.235. On this axis the diagnostic is the least powerful of the four conditional pre-tests, not the most, and its contrast slope is materially biased, at -0.111 to -0.141 against true values of 0.20 to 0.80, whereas Callaway–Sant’Anna’s contrast is unbiased to three decimals. The subgroup contrast should therefore be reported as a diagnostic that exists rather than as one that outperforms; the aggregate slope rule is where the DiD-BCF diagnostic is strong.

5.4 Tying detection to consequence

Panel (c) of Figure 2, tabulated in `Results/aggregated/pretrend_all.csv`, reports the ATT bias the violation actually induces, which is what a pre-test is ultimately protecting against. At a slope of 0.05 the plain fit’s ATT bias is 0.194 and the diagnostic detects the violation in 0.220 of replications; at 0.10 the bias is 0.393 and detection is 0.520; at 0.20, 0.787 and 0.935; at 0.40, 1.575 and 1.000. Detection therefore lags consequence at the low end, and the honest statement is that a violation large enough to bias the ATT by about a fifth of a unit will be missed four times in five by the marginal event study and about four times in five by the diagnostic as well. The posterior correction halves the induced bias throughout, at 0.109, 0.230, 0.474 and 0.960 against the plain fit’s figures, which is a further argument for the correction in designs where a residual violation is plausible.

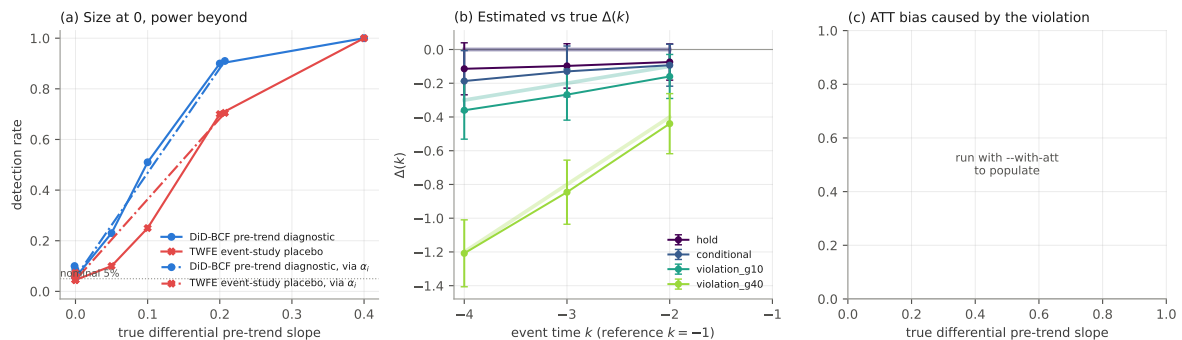


Figure 3: The same three panels at linearity degree $d = 3$. The nonlinear nuisance axis leaves the diagnostic’s size, power and discrimination essentially unchanged.

The degree axis does not disturb any of this. At $d = 3$ size is 0.055, the power curve reads 0.230, 0.510, 0.900 and 1.000, and discrimination is 0.100 against the event study’s 0.755. Every

Table 4: The same operating characteristics at $d = 3$. `PT_violation_het20` is over 150 replications; every other row is over 200.

Scenario	Violation	True slope	Detection, slope rule		Any- k Diagnostic	Slope recovery	
			Diagnostic	TWFE ES		bias	cover. 95%
<code>PT_violation_het40</code>	via α_i	-0.001	0.100	0.065	0.090	0.048	0.94
<code>PT_violation_het10</code>	via α_i	-0.000	0.055	0.045	0.030	0.044	0.94
<code>PT_violation_het20</code>	via α_i	-0.000	0.087	0.073	0.047	0.045	0.93
<code>PT_violation_a20</code>	via α_i	0.207	0.910	0.705	0.840	0.019	0.97
<code>PT_conditional</code>	unconditional only	0.000	0.100	0.755	0.055	0.065	0.94
<code>PT_violation_g05</code>	group slope	0.050	0.230	0.100	0.110	0.034	0.95
<code>PT_violation_g10</code>	group slope	0.100	0.510	0.250	0.385	0.027	0.96
<code>PT_violation_g20</code>	group slope	0.200	0.900	0.700	0.845	0.018	0.96
<code>PT_violation_g40</code>	group slope	0.400	1.000	1.000	1.000	0.011	0.95
<code>PT_hold</code>	none (PTA holds)	0.000	0.055	0.045	0.045	0.044	0.95

figure is within Monte-Carlo error of its $d = 1$ counterpart, which is a useful robustness statement given how sensitive the corrected ATT estimator is to the same axis.

6 Already-treated contamination

6.1 What the ramp shows

Table 5 and Figure 4 report mean absolute error and coverage against the measured Goodman–Bacon weight on already-treated comparisons, in the three estimand panels described above.

TWFE behaves exactly as the Goodman–Bacon decomposition predicts. Its pooled ATT absolute bias rises monotonically from 0.261 at a weight of 0.077 through 0.468 at 0.116 to 1.181 at 0.250, and its coverage collapses from 0.885 to 0.690, 0.500, 0.240, 0.005 and 0.000. On the event-study panel the same pattern is stronger still, from 0.183 to 1.383 with coverage from 0.970 to 0.238. Against a true pooled effect of about 5.3 the largest bias is roughly 22 percent. This is the reviewer’s question answered directly and it fully justifies excluding pooled TWFE as the headline benchmark for heterogeneous staggered designs.

Synthetic DiD is a different failure. Its pooled ATT absolute bias is 1.247 at the lowest weight and 1.314 at the highest, essentially flat, with coverage of 0.000 almost throughout. The error is not contamination; the estimator is targeting the weighted estimand its own weights define, which in a design with cohort effect multipliers (1.0, 1.5, 2.0) is not the equally weighted ATT used as the truth here. It should be read as an estimand mismatch and not cited as evidence about already-treated comparisons.

The three estimators built to eliminate the contamination do so. On the cohort-time panel Callaway–Sant’Anna’s mean absolute error runs 0.013, 0.014, 0.012, 0.017, 0.017, 0.033 and grf-DiD’s runs 0.021, 0.021, 0.022, 0.028, 0.012, 0.016; on the event-study panel Gardner’s runs 0.010 to 0.008. All three are flat in the weight, as they should be, since none uses an already-treated unit as a control. DoubleML is the exception among the covariate-adaptive benchmarks: its error rises from 0.023 to 0.200 over the axis, which is a real degradation, though an order of magnitude smaller than TWFE’s.

6.2 Where DiD-BCF stands

DiD-BCF is much better than TWFE and clearly worse than the purpose-built estimators. Its pooled ATT absolute bias rises from 0.003 to 0.247 (plain) and from 0.007 to 0.149 (corrected) across the four weights at which it was run, against TWFE’s 0.261 to 1.181 over the same span. At the highest weight the plain fit’s pooled error is a fifth of TWFE’s on the same panel; the purpose-built estimators sit near 0.03 on their own panels, an order of magnitude below it,

though that last comparison crosses estimands and should be read as indicative. Coverage tells the same story more sharply: the plain fit’s pooled ATT coverage falls from 0.975 to 0.850 and 0.670, while the corrected fit holds 0.975 or above at every weight. This is the second design family, after the serial sweep, in which the correction is doing real inferential work.

Two further observations belong in the write-up. First, on the cohort-time panel the plain fit’s mean absolute error is roughly 0.25 at *every* weight, including the lowest, with coverage near 0.73. That is not contamination; it is the flat, weight-independent cell-level recentring already documented for the plain specification, and the ramp is not the place to diagnose it. The corrected fit removes it at low weight, at 0.024, and loses it again at high weight, at 0.339. Second, the corrected fit is unstable on this design family at the nonlinear degrees, with a pooled ATT error of 0.384 at a weight of 0.100 and $d = 2$ against 0.050 at 0.077, which is the same instability seen at small N in Section 4 and points to the cross-fitted pilot rather than to the design.

Table 5: Mean absolute error against the measured Goodman–Bacon weight on already-treated comparisons, by natively reported estimand, at $d = 1$. 200 replications per design.

Estimand	Estimator	Weight on already-treated comparisons					
		0.077	0.100	0.116	0.136	0.196	0.250
ATT	DiD-BCF (plain)	0.003	0.015	–	–	0.119	0.247
	DiD-BCF (corrected)	0.007	0.014	–	–	0.044	0.149
	TWFE	0.261	0.386	0.468	0.573	0.889	1.181
	Synthetic DiD	1.247	1.269	1.280	1.290	1.300	1.314
GATT	DiD-BCF (plain)	0.251	0.253	–	–	0.294	0.357
	DiD-BCF (corrected)	0.024	0.035	–	–	0.113	0.339
	Callaway–Sant’Anna	0.013	0.014	0.012	0.017	0.017	0.033 [†]
	DoubleML DR-DiD	0.023	0.023	0.017	0.021	0.086	0.200 [†]
	grf-DiD (Wang)	0.021	0.021	0.022	0.028	0.012	0.016
ES	DiD-BCF (plain)	0.100	0.113	–	–	0.242	0.379
	DiD-BCF (corrected)	0.011	0.012	–	–	0.076	0.258
	TWFE	0.183	0.309	0.396	0.507	0.881	1.383
	Gardner (did2s)	0.010	0.013	0.012	0.009	0.006	0.008

[†]Computed on the replications in which the estimator returned an estimate: Callaway–Sant’Anna retains 92% of replications at this design, so the figure is a selected subsample and is not comparable with estimators that ran everywhere.

6.3 Estimability along the ramp

Callaway–Sant’Anna’s estimability degrades along this axis, from a retention of 1.00 at the lowest weight to 0.92 at the highest at $d = 1$, and to 0.785 at $d = 2$ and $d = 3$. DoubleML retains 0.925 at the highest weight. The two cells so affected are marked in Table 5, because a mean error computed on the 78.5 percent of replications in which the estimator converged is a selected subsample and is not comparable with an estimator that ran everywhere. That the flattest error curve in the table belongs to the estimator with the highest failure rate at the hardest design is itself worth stating.

7 Interpretation and remaining work

Three claims are now supported by completed evidence and can go into the revision as they stand. The first is that the plain structured DiD-BCF fit dominates every benchmark on individual-level heterogeneity recovery at every panel size in the sweep, including $N = 50$, by a factor of two against the best benchmark and a factor of five or more against every cohort-time method.

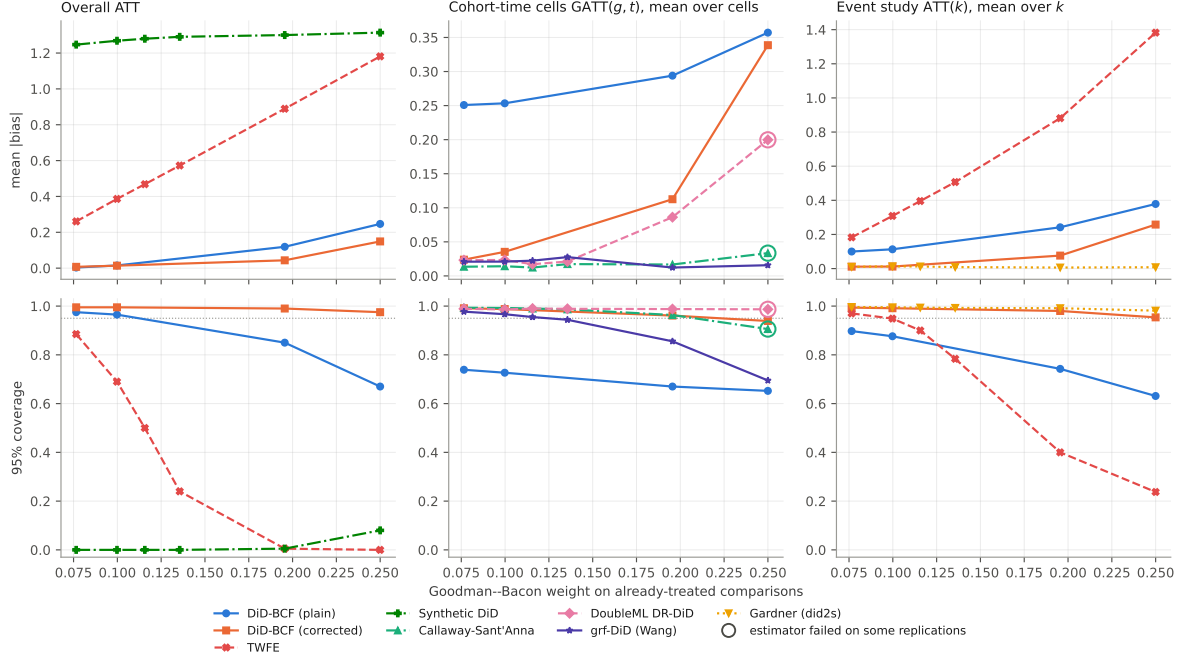


Figure 4: Mean absolute error and 95% coverage against the measured Goodman–Bacon weight on already-treated comparisons, by natively reported estimand, at $d = 1$.

The second is that the proposed pre-trend diagnostic is correctly sized, more powerful than the standard-practice marginal event-study placebo at every violation magnitude, and correctly quiet in the design where conditional parallel trends holds but unconditional parallel trends does not. The third is that pooled TWFE is unusable as a benchmark in staggered designs with a small never-treated pool, with bias up to 22 percent of the true effect and coverage of zero.

Three claims are not supported and should not be made. Pre-trend discrimination is not unique to the diagnostic: Callaway–Sant’Anna’s and grf-DiD’s conditional pre-tests do the same job, and the diagnostic’s advantage over them on the power curve is on the order of fifteen points, not an order of magnitude. The subgroup contrast is the weakest of the four conditional pre-tests on the heterogeneous-violation designs and its contrast slope is biased, so it should be presented as available rather than as superior. And DiD-BCF is not robust to already-treated contamination in the way Callaway–Sant’Anna, Gardner and grf-DiD are: its error grows by two orders of magnitude across the weight axis, from 0.003 to 0.247, even though it stays far below TWFE throughout.

The posterior correction now has a consistent profile across three design families. It repairs the plain fit’s interval calibration wherever that calibration is broken, which is the serial sweep at every N , the small panels, and the high-contamination end of the ramp, and it does so at roughly double the RMSE. It is unstable at the nonlinear degrees whenever the effective sample is small, at $N \leq 400$ in the sweeps and at the interior weights of the ramp, and it discards the individual-level heterogeneity that is the plain fit’s main advantage. The recommendation that follows is to report both specifications side by side rather than to choose, with the plain fit carrying the heterogeneity results and the corrected fit carrying the pooled inference.

Four gaps remain. The DiD-BCF fits for `D_ramp_nt20` and `D_ramp_nt25` have not been run, so the ramp shows DiD-BCF at four of six weights. One block of fifty replications is missing from `PT_violation_het20` at $d = 3$. The `PT_*` designs carry DiD-BCF ATT results at $d = 1$ only, so the detection-versus-consequence comparison of Section 5.5 has no $d = 3$ counterpart. And TWFE was not run on the sweeps below $N = 200$, so it is absent from the two smallest columns of Table 1.